Amadeo Huerta Moncho

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AI/ML researcher and engineer with a background in probabilistic machine learning, computer vision, and scalable AI systems. Experience developing reproducible ML pipelines across research and applied settings, with interest in robust machine learning, representation learning, and intelligent product development**.**

Education

MSC IN INTELLIGENT INTERACTIVE SYSTEMS | 2026 | UNIVERSITAT POMPEU FABRA

- Specialization: Probabilistic Machine Learning, Self-supervised Learning (SSL), Computer Vision

BSC IN ARTIFICIAL INTELLIGENCE | 2021-2025 | UNIVERSITAT POLITÈCNICA DE CATALUNYA

- Specialization: Machine learning, computer vision, data-driven systems
- Related coursework: Deep Learning, Data Science, Robotics, Optimization, Statistical Modelling
- Erasmus Exchange: University of Southern Denmark (SDU), 2024

Skills

APPLIED AI/ML RESEARCH

- Machine Learning research, Self-supervised Learning, Bayesian methods, Uncertainty Quantification
- Computer Vision and multimodal perception
- Experimental design, model evaluation, and research-to-engineering translation

TECHNICAL STACK

- Languages: Python, C/C++, MATLAB, SQL, R | Frameworks: PyTorch, TensorFlow, OpenCV | Systems: Spark, Hadoop, ROS/ROS2
- Reinforcement Learning (HER, RLHF, Offline RL), Large Language Models

COMMUNICATION

- Experience presenting research projects and documenting reproducible AI/ML pipelines

Experience

RESEARCH ASSISTANT | AI & ML GROUP, UNIVERSITAT POMPEU FABRA | MAR 2026 – PRESENT

- Develop ML models and information-theoretic analyses for understanding representation learning.

MEDICAL VISION RESEARCHER (INTERNSHIP) | (IOC-UPC) | JAN 2025 – JUL 2025

- Developed computer vision pipelines for radiographic analysis, including segmentation and quantitative assessment.
- Combined deep learning and classical vision methods for robust medical image processing.

SELECTED PROJECT: Autonomous navigation robot: developed real-time vision and control software for robotic navigation using ROS and Turtlebot3.

Additional Information

- Languages: Spanish (Native), Catalan (Native), English (Advanced), French (Basic)
- Certification: Research Ethics and Personal Data Protection — CIREP (UPF, 2025)